

Addressing Mode

Computer Architecture and Organization

Addressing Modes

- The operation field of an instruction specifies the operation to be performed. This operation must be executed on some data stored in computer registers or memory words.
- The way the operands are chosen during program execution is dependent on the addressing mode of the instruction.
- The addressing mode specifies a rule for interpreting or modifying the address field of the instruction before the operand is actually referenced.

Program Counter

- The PC register keeps track of the instructions in the program stored in memory.
- PC holds the address of the instruction to be executed *next* and is incremented each time an instruction is fetched from memory.

Categories of Addressing Mode

The most well known addressing mode are:

- Implied Addressing Mode.
- Immediate Addressing Mode
- Register Addressing Mode
- Register Indirect Addressing Mode
- Auto-increment or Auto-decrement Addressing Mode
- Direct Addressing Mode • Indirect Addressing Mode
- Displacement Address Addressing Mode
- Relative Addressing Mode • Index Addressing Mode

Example to be used
in the following
sections :-

PC=200

R1=400

XR=100

AC

Memory	
Address	
200	Load to AC Mode
201	Address=500
202	Next instruction
399	450
400	700
500	800
600	900
702	325
800	300

Direct Address

- It is the address (in memory) of the data to be used: *"address of the operand"*.
- Effective address is 500 and the operand to be loaded into AC is 800.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Indirect Address

- It is the address(in memory) of the *address of the data* to be used.
- Effective address is 800 and the operand is 300.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Register Mode

- Operands are in registers that reside within the CPU.

- R1=400

Simply, the operand is in R1 and 400 is loaded into AC.

Address	Memory
200	Load to AC
201	Mode
202	Address=500
	Next instruction
399	450
400	700
500	800
600	900
702	325
800	300

Register Indirect Mode

- In this mode, the selected register contains the address of the operand rather than the operand itself.

- R1=400

Effective address is 400, equal to the content of R1 and the operand loaded into AC is 700.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Immediate Mode

- In this, operand is specified in the instruction itself. It has an operand field rather than the address field.
- Operand field contains the actual operand to be used in conjunction with the operation specified in the instruction.
- The second part of the instruction is taken as an operand rather than an address

So, 500 is loaded into AC.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Auto increment Mode

- This is similar to register indirect mode except that the register is incremented after its value is used to access memory.
- R1=400, R1 is incremented i.e. R1=401 after the execution of the instruction.

Operand=700

Address	Memory
200	Load to AC Mode
201	Address=500
202	Next instruction
399	450
400	700
500	800
600	900
702	325
800	300

Auto decrement Mode

- In this mode, the register is decremented before its value is used to access memory.

- R1=400, R1 is decremented i.e. R1=399

The operand loaded into AC is now 450.

Address

Memory

200

201

202

399

400

500

600

702

800

Load to AC

Mode

Address=500

Next instruction

450

700

800

900

325

300

Relative Addressing Mode

- In this mode, the content of PC is added to the address part of the instruction in order to obtain the effective address.
- The effective address is $500+202=702$ and the operand is 325.

Address	Memory	
200	Load to AC	Mode
201	Address=500	
202	Next instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

Indexed Addressing Mode

- In this mode the content of an index register is added to the address part of the instruction to obtain the effective address.

•XR=100

$XR+500=100+500=600$

Operand= 900

Address	Memory
200	Load to AC
201	Address=500
202	Next instruction
399	450
400	700
500	800
600	900
702	325
800	300

Sample Question

- Registers $R1$ and $R2$ of a computer contain the decimal values 1200 and 5600. What is the effective address of the memory operand in each of the following instructions?
- Load $20(R1), R5$
- Move $\#3000, R5$
- Store $R2, R5$
- Add $-(R2), R5$
- Subtract $(R1)+, R5$
- Add $R5, 30(R1, R2)$

Thank You!